FINANCIAL RATIOS IN CORPORATE CREDIT ANALYSIS

Learning Outcome

- calculate and interpret financial ratios used in credit analysis

The use of financial ratios based upon the quantitative factors outlined in the prior lesson allows credit analysts to assess the relative financial health of a company, identify trends over time, and compare companies within and across industries with a focus on profitability, coverage, and leverage. Common financial ratios used to evaluate a corporate issuer's credit risk are outlined in Exhibit 3.

Exhibit 3:

Key Financial Ratios for Corporate Creditworthiness

Profitability

EBIT Margin:

$$\frac{\text{Operating Income}}{\text{Revenue}}$$

Coverage

EBIT to Interest Expense:

$$\frac{\text{Operating Income}}{\text{Interest Expense}}$$

Leverage

Debt to EBITDA:

$$\frac{\text{Debt}}{\text{EBITDA}}$$

RCF to Net Debt:

$$\frac{\text{Retained Cash Flow (RCF)}}{\text{Debt} - \text{Cash and Marketable Securities}}$$

In general, credit risk analysis using profitability, coverage, and leverage ratios seeks to establish a firm's ability to service cash flows from operating activities, as opposed to asset disposals or financing activities. Cash flow measures used in credit analysis, such as free cash flow (FCF, or net income less working capital increases, investment in fixed assets, and net interest paid), funds from operations (FFO, or net income from continuing operations, depreciation and amortization, deferred income tax, and other non-cash items), or retained cash flow (RCF, or net cash flow from operating activities less dividends paid) are more conservative than others because they make certain adjustments for cash used in managing and maintaining the business or distributed to shareholders. Debt and interest expense measures used in credit analysis also may be adjusted for operating leases, debt-like fixed lease payments, and/or other fixed off-balance-sheet obligations that impact a company's ability to service its debt.

These measures and the ratios developed based upon these measures are non-IFRS in the sense that they do not have official IFRS definitions; the concepts, names, and definitions given should be viewed as one usage among several possible, in most cases.

EBIT Margin: Credit analysts focus on EBIT (earnings before interest and taxes) to determine a company's operating performance prior to capital costs and taxes, since interest expense is paid before income taxes are calculated. In some cases, EBITDA (or earnings before interest, taxes, depreciation, and amortization expense) is used as a broader measure, which adds back (non-cash) depreciation and amortization expense. Higher EBIT margins increase profits available to service debt.

EBIT to Interest Expense: This measures the degree to which operating profit covers periodic interest payments, with a higher coverage ratio representing less credit risk. In some cases, the numerator is changed to include depreciation and amortization (EBITDA)

denominator, so care must be taken to reflect whether a greater or lesser number reflects increased leverage. For example, a higher debt-to-EBITDA ratio represents higher leverage, and debt may be adjusted to reflect operating leases or off-balance-sheet commitments. Rating agencies often use it as a trigger for rating actions, and banks reference it in loan covenants. A higher ratio indicates more leverage and thus higher credit risk. Note that this ratio can be very volatile for companies with high cash flow variability, such as those in cyclical industries and with high operating leverage (fixed costs).

Retained Cash Flow (RCF) to Net Debt: Use of cash flow rather than earnings measures is also common in measuring leverage, while here the debt measure is reduced by available cash. A higher RCF to net debt measure implies lower leverage.

While financial ratios used in credit analysis vary across industries, the values of financial ratios themselves for a given rating or level of credit quality may also range widely across industries due to different business models, competitive pressures, and other factors.

Credit analysts use financial modeling to evaluate how these ratios might change over time as revenue drivers change, as in the following example:

EXAMPLE 4

Bowstream Corporation Credit Risk Assessment

A credit analyst has created a financial statement model to evaluate the credit risk of Bowstream Corporation, a mid-sized, investment-grade consumer durables company. While Bowstream's most recent financial results show a solid 10% sales growth and a robust 35% operating profit margin, the analyst would like to assess Bowstream's performance under a downside scenario where the company's production and interest costs rise due to inflation amid slower sales growth. Model assumptions used for Bowstream's sales growth, cost of goods sold as a percentage of sales, interest rates, and debt growth over the next five years are as follows:

	Reported	Projected				
	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Sales growth	10%	5%	5%	5%	5%	5%
Cost of goods sold/Sales	65%	70%	75%	80%	85%	85%
Interest rate on debt	6%	7%	7%	7%	8%	8%
Debt growth rate	10%	10%	10%	10%	10%	10%

Selected financial data and projections from these three-statement model results based on these assumptions and stable ratios of balance sheet and other positions to sales are as follows:

Income Statement	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	**5,225**	**5,486**	**5,761**	**6,049**	**6,351**	**6,669**
Cost of goods sold	(3,396)	(3,840)	(4,320)	(4,839)	(5,398)	(5,668)
Gross profit	**1,829**	**1,646**	**1,440**	**1,210**	**953**	**1,000**
Gross profit (%)	35%	30%	25%	20%	15%	15%
Selling, general, and administrative	(499)	(524)	(550)	(577)	(606)	(637)
Interest payments on debt	(38)	(46)	(51)	(56)	(72)	(83)
Interest earned on cash and mktable sec	12	12	12	12	12	12
Depreciation	(400)	(467)	(517)	(547)	(559)	(552)
Profit before tax	**905**	**621**	**335**	**41**	**(273)**	**(259)**

Net income (% of sales)	11%	7%	4%	0%	−3%	−3%
Dividends	(118)	(81)	(44)	(5)	—	—
Retained earnings	**470**	**323**	**174**	**21**	**(177)**	**(168)**

Balance Sheet	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Cash and marketable securities	400	400	400	400	400	400
Current assets	627	658	691	726	762	800
Fixed assets						
Cost basis	5,993	6,840	7,594	8,233	8,735	9,256
Accumulated depreciation	(2,400)	(2,867)	(3,383)	(3,931)	(4,490)	(5,042)
Net fixed assets	3,593	3,973	4,211	4,303	4,244	4,214
Total assets:	**4,620**	**5,032**	**5,302**	**5,428**	**5,407**	**5,414**
Current liabilities	523	549	576	605	635	667
Debt	627	690	759	835	960	1,104
Stock (225 shares issued @ 10)	2,250	2,250	2,250	2,250	2,250	2,250
Accumulated retained earnings	1,220	1,543	1,718	1,739	1,562	1,394
Total liabilities and equity	**4,620**	**5,032**	**5,302**	**5,428**	**5,407**	**5,414**

Consolidated Statement of Cash Flows	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Net cash from operating activities	978	866	729	568	376	377
Net cash used in investing activities	(918)	(847)	(754)	(639)	(501)	(521)
Net cash from financing activities	(61)	(18)	25	71	125	144

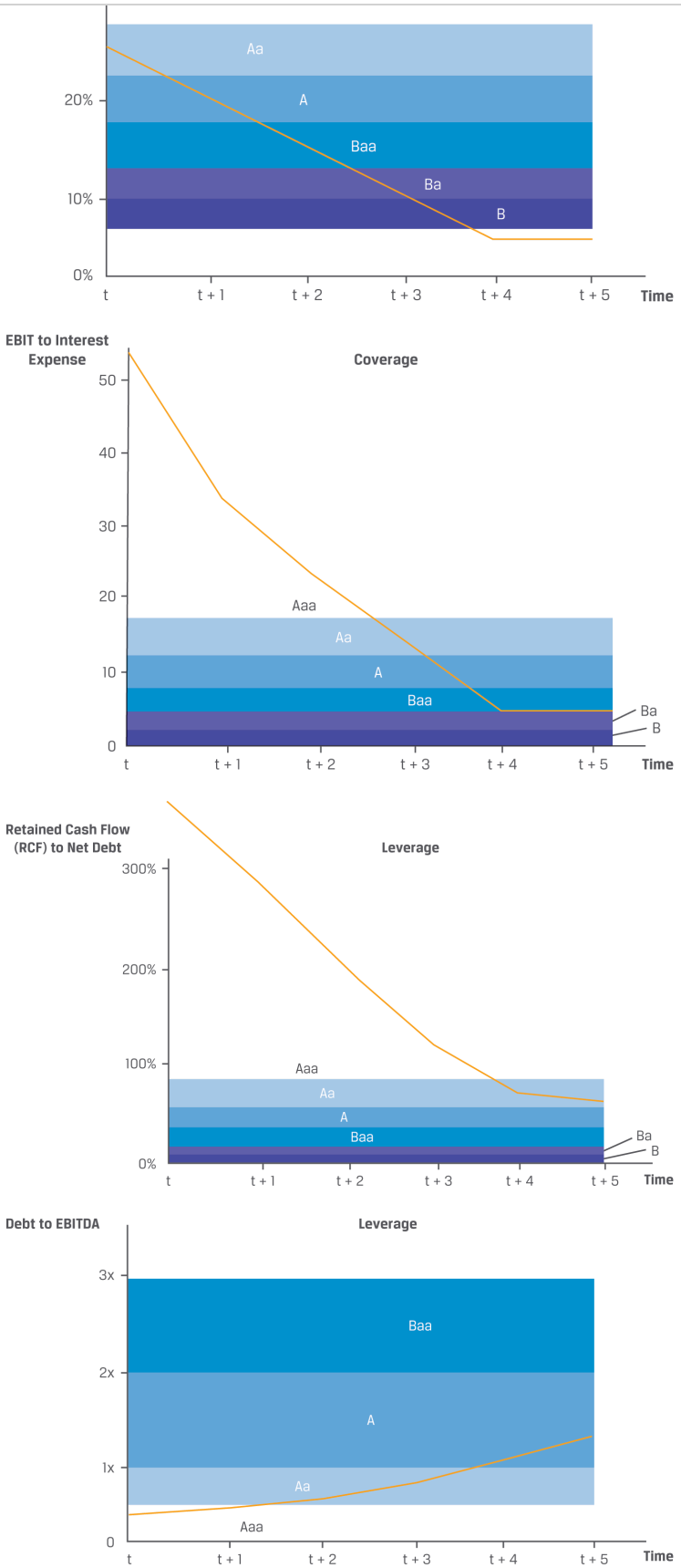
The analyst uses data from these three-statement model results to solve for recent and projected profitability, coverage, and leverage ratios in Exhibit 3:

	Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
EBIT	930	655	373	86	-212	-189
EBITDA	1,330	1,122	890	633	347	363
FFO (Net cash from operating activities)	978	866	729	568	376	377
RCF (FFO − Dividends)	861	785	685	563	376	377
EBIT margin	25%	20%	15%	10%	5%	5%
EBIT to Interest expense	51.9	32.9	23.0	14.4	5.8	5.2
Debt to EBITDA	0.47	0.61	0.85	1.32	2.77	3.04
RCF to Net debt	379%	271%	191%	130%	67%	54%

Under this scenario, Bowstream faces declining profitability (as EBIT margin falls), weaker debt coverage (lower EBIT-to-interest expense), and increased leverage (a higher debt to EBITDA and lower RCF to net debt), which is likely to lead to wider credit spreads and a potential ratings downgrade.

In order to measure these changes, analysts often compare existing and projected ratios to industry and credit rating peers to determine expected future credit quality changes as categorized by expected rating. For example, consider the following comparisons based upon Moody's ratings methodology and these ratios for consumer durables:

Figure 1



Source: Moody's Investors Service, "Methodology Announcement: Moody's Updates Its Methodology for Rating Consumer Durables Companies" (10 September 2021).

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Question Set

B. EBITDA/Interest Expense
C. Debt/Capital
D. EBITDA
E. Debt/EBITDA
F. FCF
G. EBIT/Interest Expense

1. Profitability and cash flow metrics
2. Leverage metrics
3. Coverage metrics

Hide Solution

Solution:

1. Profitability and cash flow metrics: D, F
2. Leverage metrics: A, C, E
3. Coverage metrics: B, G

2. State true or false to the following statement and explain your reasoning: The higher the coverage ratios, the higher the credit risk of the borrower.

Hide Solution

Solution:

False. Coverage ratios measure an issuer's ability to cover its interest payments. The higher the coverage ratios, the *lower* the credit risk of the borrower.

3. Calculate FFO/debt based on the following financial information:

Net income from continuing operations	180
Depreciation & amortization	30
Deferred income taxes	18
Other non-cash items	28
Increase in working capital	(52)
Dividend paid	(60)
Capital expenditure	(100)
Debt	1,280

Hide Solution

Solution:

FFO = Net income from continuing operations + Depreciation & amortization + Deferred income taxes + Other non-cash items = 180 + 30 + 18 + 28 = 256.

FFO/debt = 256/1,280 = 0.2 or 20%.

4. Company X and Company Y operate in the same industry. Based on the financial information below, which company has lower credit risk?

	Company X	Company Y
EBITDA margin	15%	15%
FCF	(20)	20
FCF before dividends	(45)	18
Debt/EBITDA	2.5	1.6
EBITDA/interest expense	3.0	3.5

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same EBITDA margin, Company Y has positive FCF and FCF before dividends, a lower leverage ratio, and a higher coverage ratio than Company X.

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Discussion

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PZ in Example4 , why is the EBIT 1,330 in Year 0? it should be EBIT = Profit Before Tax + Interest Payments - Interest Earned (EBIT = 905 + 38 − 12 = 931)?

Created 9 months ago by PIERFRANCESCO ZEOLI 1 reply | Last Activity: 4 months ago

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